

Audio Groove

Axiom Audio Groove · Constitutional Audio Intelligence · Compact Sensory Representation

"A physical record stores sound as groove geometry. Axiom stores it the same way — depth for energy, width for spread, curve for rhythm — so a micro-agent can reason about sound without hauling the whole thundercloud."

Concept Note · Antonio Roberts · Orivael · June 2026

Status: CONCEPT — Software implemented (axiom_audio/)	Pairs with: ORVL-025 (Event Token) · ORVL-004 (MKB)	Domain: Audio · Voice · Automotive
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Abstract.

Axiom Audio Groove converts raw PCM audio into a compact groove-geometry profile — depth (low-frequency energy / loudness), width (stereo spread / dynamic range), curve (pitch motion / rhythm pattern), and texture (noise floor / breath presence). Specialist micro-agents each report one property; a coordinator combines them into a signed AUDIO_GROOVE_BLOCK. The block is inspectable, routable, verifiable, and evidence-backed — it stores the signal meaning without dragging raw samples forward. The existing implementation in axiom_audio/ ships: AmbientAudioAgent (material + impact + decay classification), VoiceActivityDetector, BPM tempo estimator, voice characterizer, and automotive adapter (ENGINE/BRAKE/CABIN/VOICE) — all producing HMAC-signed reports under axiom-audio-v1.

01 | AUDIO AS GROOVE GEOMETRY

1. Audio as Groove Geometry

A vinyl record stores sound as tiny groove variations. The AI version calculates digital sound-shape features. Those measurements create a compact audio profile that any agent can inspect without decoding the original waveform.

Audio Property	Groove-style Feature	AI Meaning
Volume / loudness	Depth / amplitude	Energy, force, emphasis
Bass / low-end	Wider or heavier motion	Weight, body, rumble
Treble / detail	Fine surface texture	Sharpness, clarity, hiss
Rhythm / cadence	Repeated spacing patterns	Timing, groove, hesitation

Audio Property	Groove-style Feature	AI Meaning
Pitch movement	Curve direction	Question, confidence, tension
Stereo / location	Groove width / spread	Spatial placement

02 | AUDIO MICRO-AGENT STACK

2. Audio Micro-Agent Stack

The audio property map is split into specialist agents. Each agent listens for one type of signal, produces a compact signed report, and passes it to a coordinator. This makes the system modular instead of forcing one large model to understand every audio detail at once.

Agent	Listens For	Example Output
Depth Agent	Low-end energy, loudness, pressure	Voice has strong low-mid body
Width Agent	Stereo spread and placement	Sound is narrow and centered
Pitch Agent	Rising / falling tone	Pitch rises at end — likely question
Rhythm Agent	Timing, beat, pauses, cadence	Speaker hesitates before key words
Texture Agent	Noise, breath, distortion, grain	Slight room hiss and breath presence
Emotion Agent	Stress, calm, excitement, uncertainty	Delivery suggests uncertainty
Intent Agent	Delivery meaning and communication role	Speaker is asking, not commanding

Coordinator

The coordinator receives all seven micro-agent reports and merges them into a single `AUDIO_GROOVE_BLOCK` with an `evidence_trace` listing which agents contributed. Any downstream system can verify the block without re-running the agents.

03 | AUDIO_GROOVE_BLOCK FORMAT

3. AUDIO_GROOVE_BLOCK Format

Each audio analysis result is serialised as a compact block. The block carries groove geometry fields, a confidence score, an evidence trace, and an HMAC-SHA256 signature over the canonical JSON form under the `axiom-audio-v1` key namespace. Signature failure refuses the block at any inspection point.

```
AUDIO_GROOVE_BLOCK
source: voice_sample_001
duration: 4.2s
groove_depth: { low_freq_energy: 0.74, vocal_weight: 0.62 }
groove_width: { stereo_spread: 0.31, dynamic_range: 0.48 }
groove_curve: { pitch_motion: rising, rhythm_pattern: syncopated }
```

```
texture: { noise_floor: low, breath_presence: medium }
meaning: { likely_intent: questioning, confidence: 0.81 }
evidence_trace: [depth_agent, width_agent, pitch_agent, rhythm_agent,
texture_agent, emotion_agent, intent_agent]
signature: <HMAC-SHA256 over canonical JSON, axiom-audio-v1>
```

Field	Content
groove_depth	Low-frequency energy ratio (depth) and vocal weight proxy
groove_width	Stereo spread and dynamic range (amplitude envelope width)
groove_curve	Pitch motion direction and rhythm pattern classification
texture	Noise floor level and breath / grain presence
meaning	Likely intent class and confidence produced by Intent Agent
evidence_trace	Ordered list of contributing agent names — the audit chain
signature	HMAC-SHA256 under axiom-audio-v1; verified before any downstream use

4. Implemented Agents (axiom_audio/)

The `axiom_audio/` package ships a Phase A + B implementation. Five modules cover the seven micro-agent roles described above. Each produces a signed report compatible with the ORVL-025 Event Token Audio layer.

Module	Agent Role	Key Output
ambient.py	Depth + Texture + Rhythm + Material	impact_profile, material_signature (glass-like / metal-like / wood-like / fabric-like), decay_pattern, depth, width, rhythm — HMAC-signed AudioReport
vad.py	Width (activity detection)	activity_ratio, voiced_regions — VoiceActivityDetector signs VADReport
voice.py	Pitch + Emotion	pitch characterization, speech detection — Phase B (no ASR / no speaker ID)
tempo.py	Rhythm (BPM)	Estimated beats-per-minute from onset timing
automotive.py	Coordinator (KIA US adapter)	Routes to ENGINE / BRAKE / CABIN / VOICE / SILENCE based on material + width + VAD — AutomotiveAudioEvent with severity (NORMAL / WARNING / CRITICAL)

Automotive safety example

brake squeal: material_signature=metal-like + rhythm=periodic + width > 0.80 → BRAKE CRITICAL immediately, regardless of VAD state. The evidence trace shows which threshold triggered the alert, satisfying the explainability requirement.

5. Cross-Patent Wiring

Patent	Role for Audio Groove
ORVL-004 MKB	Each micro-agent report can be registered as a KnowledgeBlock (<code>block_type=AUDIO_GROOVE</code>). The coordinator's merged block is the composed output certified by CBV before downstream use.

Patent	Role for Audio Groove
ORVL-010 CBV	Groove geometry thresholds (SILENCE_FLOOR, SHARP_ATTACK_MS, decay limits) are CANNOT_MUTATE in the module; CBV certifies the constraint set before the agent enters the certified registry.
ORVL-022 CPI	AmbientAudioAgent's material_signature (glass-like / metal-like) feeds the CPI VertexClassifier's fracture_probability override — audio evidence influences physical grip decisions.
ORVL-025 Event Token	AudioReport is the Audio layer of an EventToken. groove_depth / groove_width / groove_curve / texture / meaning map directly to the EventToken Audio payload schema.
ORVL-027 Video Topology	Audio groove (timing, rhythm) and video topology (object motion, events) merge into a single multimodal evidence graph — audio gives tone and intent, video gives physical cause.

6. Provisional Patent Claims (ORVL-026)

Claim 1. A method of representing audio as groove geometry wherein a raw PCM signal is analysed to produce six named groove fields — depth (low-frequency energy), width (spectral spread), curve (pitch-motion direction), texture (noise and breath presence), rhythm (onset-pattern classification), and meaning (intent confidence) — and the complete field set is serialised as a compact AUDIO_GROOVE_BLOCK with an HMAC-SHA256 signature over its canonical JSON form, such that downstream agents can inspect, route, and verify the block without decoding the original waveform.

Claim 2. A modular micro-agent architecture for audio analysis wherein each of seven specialist agents — Depth, Width, Pitch, Rhythm, Texture, Emotion, and Intent — independently processes one acoustic property of the input signal, produces a compact signed sub-report, and passes it to a coordinator that merges the reports into a single AUDIO_GROOVE_BLOCK carrying an evidence_trace listing the contributing agents.

Claim 3. A material-signature classifier operating on spectral centroid and high-frequency energy ratio that maps audio to one of four constitutional material classes — glass-like (centroid > 4 kHz, hf_ratio > 0.55), metal-like (centroid > 3 kHz with resonant peak), wood-like (800–2500 Hz, hf_ratio < 0.4), or fabric-like (centroid < 800 Hz, hf_ratio < 0.2) — and exposes the classified material class as a constitutional input to a physical-intelligence grip-planning pipeline (ORVL-022 CPI).

Claim 4. An automotive audio safety adapter that classifies in-cabin PCM audio into the event types ENGINE, BRAKE, CABIN, VOICE, and SILENCE using only the groove geometry fields of the AUDIO_GROOVE_BLOCK — specifically width (spectral spread as brake-squeal proxy), material_signature, rhythm, and VAD activity_ratio — and emits a severity verdict (NORMAL / WARNING / CRITICAL) with an evidence trace identifying which thresholds triggered the classification, without requiring a neural network or cloud round-trip.

Claim 5. A groove-geometry signing protocol wherein each AudioReport is HMAC-signed under a namespace key derived exclusively for audio (axiom-audio-v1) — distinct from the Event Token layer key (axiom-event-token-layer-v1) and all other signing surfaces — so that an audio report can be verified standalone and cross-namespace replay cannot move a forged audio payload into a different signing context.

Claim 6. A cross-modal evidence composition method wherein an AUDIO_GROOVE_BLOCK (produced by ORVL-026) and a VIDEO_TOPOLOGY_BLOCK (produced by ORVL-027) are merged by a multimodal coordinator into a single signed evidence graph, with audio supplying timing, tone, and intent, and video supplying objects, surface state, and physical causality, and the merged graph constituting the Audio + Video layers of an Axiom Event Token (ORVL-025).

“Store the weather map, not the thundercloud. Audio Groove gives every agent the signal meaning — depth, width, curve — without the raw waveform riding along.”